

Bridge and Local Bridge

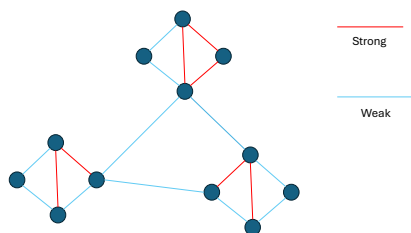
- **Bridge** and **local bridge** is another key concept.
 - ▶ (i, j) is a **bridge** if deleting it breaks one large component including i and j into two separate components (one with i and the other with j).
 - ▶ (i, j) is a **local bridge** if there is no $k \in V$ such that $(i, k) \in E$ and $(j, k) \in E$ (i and j have no common neighbor).
- A bridge is clearly a local bridge. But a local bridge may not be a bridge. If bridge (i, j) is deleted, then there would be no path between i and j . If a local bridge (i, j) is deleted, there may be still a long path between i and j .

Now observe the following.

Proposition

Suppose that G satisfies the STC property. If (i, j) is a local bridge and either i or j has at least one strong tie with some $k \neq i, j$, then (i, j) must be a weak tie.

This is true almost by definition. If not, then there must be a link between j and k by the STC property. But then (i, j) would not be a local bridge, a contradiction.



Here is an example of graph that satisfies the STC property. We have a collection of tight-knit groups that are connected by local bridges of weak ties. If a new information always comes from another group, then it must come through a weak tie.

Why may new information arrive through a weak tie? Here is one possible story (assume that all local bridges are weak ties).

- Delete all the local bridges from graph G . Then G is decomposed to multiple components.
- Assume that each component is a community with similar members in the sense that if someone in the community qualifies/does not qualify for a job, then everyone else qualifies/does not qualify for a job.
- Imagine that a new job opportunity arises at some node. Then either she takes up the job herself or no one in her community is qualified for the job. In the latter case, this opportunity is informed to some neighbor community through a local bridge. Then the gate-keeper at the bridge accepts the job, or this information is further conveyed to another community and so on....
- In this story, whenever someone hears about the job and accepts it, it is always through a weak tie.

Empirical Support for the STC Property

- Is the STC property borne out by any evidence?
- Onnela et.al. (2007) defines more granular version of the strength and “local bridgeness” for each link in some cell-phone network and checks how they are related.

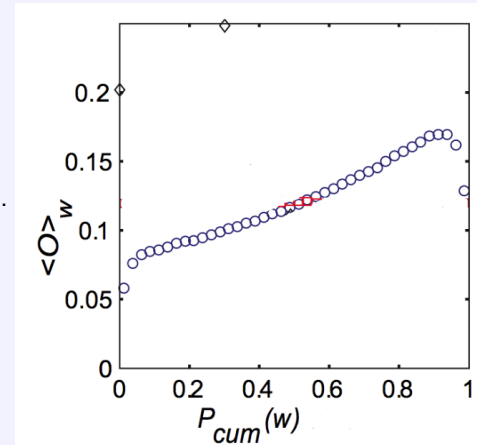
- **Neighborhood overlap** of an edge (i, j) is defined by

$$\frac{\text{number of nodes who are neighbors of both } i \text{ and } j}{\text{number of nodes who are neighbors of at least one of } i \text{ and } j}$$

(Note: the denominator does not include i and j themselves).

This index is 0 for a local bridge. We can interpret a smaller neighborhood overlap as more “local bridgeness”.

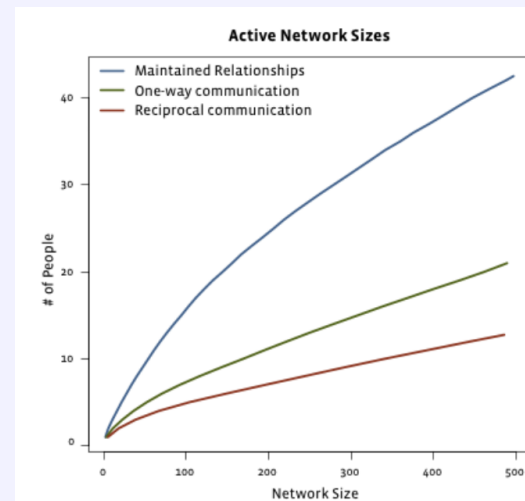
- STC property suggests that a stronger link should be associated with a larger neighborhood overlap.
- Such monotonicity in fact appears in the right plot, where x-axis represents a cumulative strength and y-axis represents a neighborhood overlap.



Scarcity of Strong Ties

- For online networks, we can keep a long list of “friends”.
- But it is costly to maintain strong ties with them. How scarce are strong ties for online networks?
- Marlow et.al. (2009) considers various type of ties for a friendship network in FB and studies how their numbers change as the total number of friends grow.
 - *reciprocal relationships*: send and receive a message
 - *one-way communication*: send a message (but may or may not receive a message)
 - *maintained relationship*: follow information (independent of messages)

- The number of ties of all three types is small relative to the total number of friends.
- The fraction of reciprocal communication is especially small and does not grow much.
- Is a maintained relationship a “weak link”?

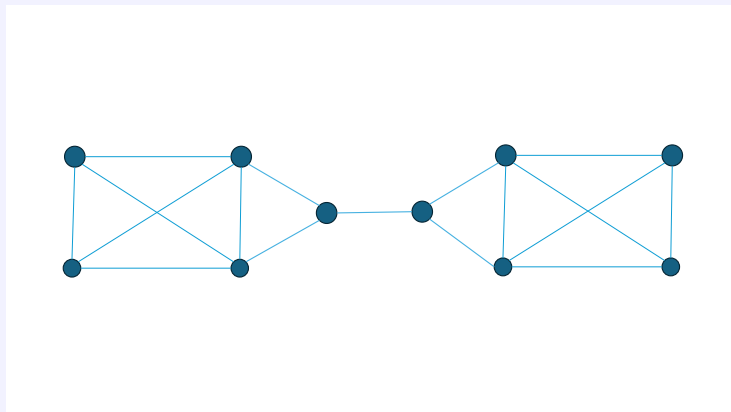


Local Bridge and Embeddedness

- (i, j) is a local bridge $\Leftrightarrow i$ and j have no common neighbor.
- Generalizing the notion of local bridge, we say that (i, j) has an **embeddedness** of K when i and j has K common neighbors. So, each local bridge corresponds to an embeddedness of 0.
- Higher embeddedness may be associated with trust within the tight-knit community they belong to. Interactions through a link of higher embeddedness is more secure. On the other hand, interactions through a local bridge may be more risky.

Structural Hole

- Local bridge may be associated with different kind of advantages:
 - ▶ Fresh information (as we already discussed)
 - ▶ Creative ideas
 - ▶ Within an organization, a local bridge fills in a **structural hole** and works as a “gate-keeper” to two groups that would be separated without the bridge.
- A person who has access to more local bridges may have some special advantages in an organization.



Summary

- Two types of links: strong ties and weak ties
- Strong Triadic Closure Property $\Rightarrow$ local bridges are weak ties
- Some empirical evidence of STC and scarcity of stronger ties.
- Some interpretation of links of smaller embeddedness (local bridge) and links of larger embeddedness.

Suggested reading. EK Ch.3.1-3.5

Who has an access to more local bridges and who has more edges of higher embeddedness?